Big Data Analysis

Final Project Assignment

Large-Scale Analysis of Mortgage Lending Data

Primary tools:	Python, Pandas, PySpark, and data visualization libraries
Dataset:	Home Mortgage Disclosure Act (HMDA), 2024
Optional extension:	Predictive analytics
Instructor:	Huan Vu, Faculty of DS&AI
Duration:	4–5 weeks

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1 Project Overview

Financial institutions generate large volumes of data through mortgage applications and lending activities. Analyzing these data can help us understand lending patterns, compare financial institutions, examine geographic differences, and identify relationships between borrower characteristics and loan outcomes.

In this project, you will use the 2024 Home Mortgage Disclosure Act (HMDA) loan-level dataset to investigate a meaningful problem in mortgage lending through large-scale data analysis.

You will develop an end-to-end analytical workflow, starting from raw data and ending with evidence-based financial insights. Your work must demonstrate data preparation, descriptive and diagnostic analysis, implementation using both Pandas and Apache Spark, comparison of analytical results, and a performance experiment.

Central objective:

> Use the 2024 HMDA dataset to discover meaningful financial patterns, investigate relationships in mortgage lending, and evaluate the correctness and performance of Pandas and Spark implementations.

Descriptive and diagnostic analysis is mandatory for every group. Predictive modeling is optional and is not required to obtain full marks.

2 Dataset

2.1 Data source

Use the 2024 HMDA loan-level data published by the Consumer Financial Protection Bureau (CFPB) and the Federal Financial Institutions Examination Council (FFIEC).

Official resources:

- HMDA data downloads: `https://ffiec.cfpb.gov/data-publication/`
- HMDA data browser: `https://ffiec.cfpb.gov/data-browser/`
- HMDA documentation: `https://ffiec.cfpb.gov/documentation/`

The required reporting year is fixed at **2024**. Do not substitute another reporting year.

2.2 Data description

HMDA contains records concerning mortgage applications and covered loans. Depending on the reporting requirements and the fields available in the 2024 file, relevant variables may include:

- Loan type and loan purpose
- Application date or reporting-period information
- Action taken / application outcome
- Loan amount
- Applicant and co-applicant characteristics
- Income and debt-to-income ratio
- Interest rate and loan costs, where reported
- Property location and occupancy information
- Financial institution identifiers

You must consult the official 2024 data dictionary and documentation. Do not assume that every possible variable is present or has the same definition across reporting years.

2.3 Initial requirements

Each group must:

1. Download or access the instructor-specified 2024 dataset.
2. Read the official documentation and data dictionary.
3. Identify the unit of observation and the meaning of important variables.
4. Record the file format, file size, number of records, and number of variables.
5. Assess whether the data support the proposed research questions.

Do not assume that every record represents a completed mortgage loan. Distinguish applications, originated loans, denials, and other reported outcomes where relevant.

3 Project Requirements

3.1 Part 1: Problem Formulation

Identify a meaningful problem in mortgage lending and formulate specific research questions that can be answered using the 2024 HMDA data.

Your proposal must include:

- The financial problem and its practical importance
- One main research question and two to four supporting questions
- The variables required to answer each question
- The planned analytical methods
- The expected financial insights

Questions must be specific, measurable, and supported by available data.
For example, “Analyze mortgage data” is too broad. A more appropriate question is:

> How do mortgage application outcomes vary across loan types, geographic areas, and financial institutions in 2024?

3.2 Part 2: Data Engineering and Preparation

Prepare the raw data for reliable analysis. Demonstrate that you understand the structure and quality of the data rather than simply loading it and producing charts.

A. Data inspection

- Examine the schema and data types.
- Count records and identify important variables.
- Inspect numerical and categorical variables.

B. Data quality assessment

- Identify missing, unavailable, or specially coded values.
- Investigate potential duplicate records.
- Identify invalid, inconsistent, or implausible values.
- Distinguish genuine outliers from data errors.

C. Data cleaning

- Apply justified rules for handling missing values.
- Correct data types and standardize representations where needed.
- Document filtering and exclusion decisions.
- Explain how these decisions may affect the analysis.

D. Feature engineering

- Create relevant numerical or categorical features.
- Derive financial indicators when appropriate.
- Explain the meaning and purpose of each derived feature.

E. Data storage

Save the prepared analytical data in an appropriate format, such as Parquet, where feasible. Explain the chosen format and any relevant partitioning decisions.

Cleaning rules must be documented and applied consistently in the Pandas and Spark implementations. Do not automatically replace every missing value with zero or remove observations solely because they appear unusual.

3.3 Part 3: Mandatory Descriptive and Diagnostic Analysis

This component is mandatory for every group.

Use the 2024 HMDA data to describe lending activity, discover patterns, and investigate relationships relevant to your research questions.

A. Descriptive statistics

Provide appropriate summary statistics for variables relevant to your project, such as:

- Number and proportion of applications by outcome
- Distribution of loan amounts
- Distribution of reported interest rates, where available
- Summary statistics for income and other financial indicators
- Counts and proportions across loan types and purposes

Choose statistics appropriate to each variable and explain why they are useful.

B. Comparative analysis

Compare lending patterns across at least two meaningful dimensions, such as:

- Loan type or loan purpose
- Financial institution
- Geographic area
- Applicant characteristics

For example, investigate whether application outcomes or loan amounts differ across loan types or geographic areas.

C. Relationship analysis

Investigate relationships between relevant variables. Examples include:

- Loan amount and applicant income
- Reported interest rates and loan characteristics
- Application outcomes and financial or demographic variables
- Differences in lending patterns across financial institutions

Use appropriate methods such as grouped comparisons, cross-tabulations, correlation analysis, or statistical summaries. An observed association does not establish causation.

D. Visualization

Include visualizations that directly address your research questions. Suitable examples include:

- Histograms and box plots for numerical distributions
- Bar charts for categorical comparisons
- Heatmaps for cross-tabulations or correlations
- Geographic visualizations, where suitable location data and tools are available

Every major visualization must have a clear title, labeled axes, appropriate units, and an interpretation. Do not include charts merely to increase the number of figures.

E. Financial interpretation

For every major finding, explain:

1. What pattern was observed?
2. What evidence supports the finding?
3. Why might the pattern matter in mortgage lending?
4. What alternative explanations or limitations should be considered?

Where the data permit, discuss implications for lenders, borrowers, financial analysts, or policymakers.

3.4 Part 4: Dual Implementation Using Pandas and PySpark

This component is mandatory for every group.

Each group must implement its core descriptive and diagnostic analyses using both Pandas and PySpark. The purpose is to compare correctness, performance, and the practical trade-offs between single-machine data analysis and distributed processing.

The two implementations must use equivalent input records, the same analytical definitions, and consistent data-cleaning rules.

A. Pandas implementation

Use Pandas to implement the main analytical workflow, including:

- Loading and inspecting the data
- Handling missing and invalid values
- Applying data-cleaning rules
- Creating derived variables
- Computing descriptive statistics
- Performing grouped aggregations and comparisons
- Producing the main analytical tables and results

If the full dataset exceeds available memory, you may use chunked processing or a justified subset for Pandas. Document the limitation and ensure that direct comparisons use equivalent input records.

B. PySpark implementation

Implement corresponding analyses using PySpark. Your implementation must include:

- Reading the data with Spark and inspecting its schema
- Applying equivalent data-cleaning rules
- Creating the same derived variables
- Performing descriptive statistics and aggregations
- Comparing relevant groups or categories
- Producing results corresponding to those from Pandas

Use appropriate Spark DataFrame operations such as `select`, `filter`, `groupBy`, `agg`, and `join`. Window functions may be used where relevant. You are not required to use every Spark API.

C. Equivalence and correctness

Compare Pandas and Spark results for at least **three important analytical outputs**. Examples include:

- Record counts after major cleaning steps
- Missing-value counts for important variables
- Mean, median, or other descriptive statistics
- Counts and proportions of application outcomes
- Average loan amounts by loan type or geographic area
- Aggregated results by financial institution

For each comparison, report:

1. The analytical quantity being compared
2. The result from Pandas
3. The result from PySpark
4. Whether the results agree
5. An explanation of any discrepancy

Results should agree within an appropriate numerical tolerance when the implementations use equivalent records and definitions. Small floating-point differences may occur because numerical aggregations can be performed in different orders.

If results disagree, investigate possible causes such as different missing-value handling, data types, filtering conditions, duplicate treatment, or aggregation definitions. Do not modify results merely to make the implementations appear identical.

3.5 Part 5: Performance and Scalability Comparison

This component is mandatory for every group.

Compare the execution performance of Pandas and PySpark on at least one common analytical workload.

Choose a representative operation, such as:

- Grouping applications by loan type and calculating summary statistics
- Computing application outcome rates by geographic area
- Aggregating loan amounts by financial institution
- Calculating descriptive statistics for numerical variables

A. Experimental design

Document:

- The operation being evaluated
- The dataset and number of records processed
- The hardware and software environment
- The Pandas and PySpark implementations

- The timing procedure
- Relevant memory or resource limitations

Both implementations must operate on equivalent input data and produce equivalent analytical results.

Where feasible, evaluate multiple data volumes, such as 25%, 50%, and 100% of the selected dataset. If Pandas cannot process the complete dataset, use the largest common feasible input for direct comparison and report the limitation.

B. Timing methodology

Measure the analytical operation separately from unrelated activities such as downloading data, installing packages, or displaying plots.

For Spark, distinguish initial setup and execution overhead from the time required for the measured workload. Document whether the Spark session was already initialized and whether data had already been cached.

Where practical, repeat the experiment and report the median execution time. Use the same machine and comparable conditions for both implementations.

C. Results and interpretation

Present a table similar to the following, replacing example data sizes with those actually tested:

Data size	Pandas (s)	PySpark (s)	Faster method
25%			
50%			
100%			

Calculate speedup when both measurements are valid:

$$\text{Speedup} = \frac{T_{\text{Pandas}}}{T_{\text{Spark}}},$$

where T_{Pandas} and T_{Spark} are the measured execution times for equivalent workloads.

Interpret the results by answering:

1. Which implementation was faster for the selected workload?
2. Did relative performance change as data volume increased?
3. What overheads affected Spark performance?
4. Did memory limitations affect either implementation?
5. Under what circumstances would Spark be preferable to Pandas?

Do not assume that Spark must be faster. On small datasets, Pandas may perform better because Spark incurs execution and coordination overhead. Conclusions must follow from measurements rather than expectations.

4 Optional Predictive Analysis

Predictive analysis is **optional**. A group can achieve full marks without developing a prediction model.

Groups choosing this extension may investigate:

- Predicting application outcomes from appropriate application characteristics
- Forecasting geographic lending activity, if sufficient temporal data are available
- Identifying unusual lending patterns using a justified analytical or machine-learning method

If you undertake predictive analysis, you must:

1. Define the prediction target and intended use.
2. Select and justify input variables.
3. Explain the model and why it is suitable.
4. Use an appropriate training and evaluation procedure.
5. Report suitable evaluation metrics.
6. Discuss limitations, data leakage, and potential bias.

For temporally ordered data, use chronological training and testing where appropriate. For application-level classification, avoid using variables that reveal the outcome or would not be available at the intended prediction time.

Use methods you can explain and justify. A complex model is not inherently better than a simple, correctly evaluated model.

The optional extension may earn up to five bonus percentage points, subject to a maximum final score of 100.

5 Financial and Ethical Considerations

HMDA data can support analysis of lending patterns and differences in application outcomes. However, interpretation requires care.

All groups must:

- Distinguish descriptive evidence from causal claims.
- Avoid treating differences in approval or denial rates as proof of discrimination.
- Acknowledge confounding factors and unavailable variables.
- Consider missing data and reporting requirements.
- Discuss the limitations of demographic comparisons.
- Avoid unsupported recommendations about individual borrowers or financial institutions.

The goal is to produce a defensible analysis, not to confirm a predetermined conclusion.

6 Project Deliverables

6.1 Deliverable 1: Project Proposal

Submit a proposal of approximately 1–2 pages before conducting the full analysis. It must contain:

1. Project title
2. Financial problem and motivation
3. Main research question and supporting questions
4. Dataset and relevant variables

5. Planned descriptive and diagnostic analyses
6. Planned Pandas and Spark implementations
7. Proposed correctness and performance comparisons
8. Optional predictive extension, if applicable

6.2 Deliverable 2: Technical Report

Submit a report of approximately 8 pages, excluding references and appendices.

Recommended structure:

1. Introduction and problem formulation
2. Dataset description and data quality
3. Data engineering and preparation
4. Descriptive and diagnostic analysis
5. Pandas implementation
6. PySpark implementation
7. Correctness comparison between Pandas and Spark
8. Performance and scalability experiment
9. Optional predictive analysis
10. Financial interpretation and limitations
11. Conclusion
12. References

Omit the optional predictive-analysis section if you do not undertake that extension. Include evidence supporting your conclusions. Place lengthy code and supplementary results in an appendix or repository.

6.3 Deliverable 3: Code and Reproducibility

Submit an organized project directory. A suggested structure is:

```
project/
|-- README.md
|-- requirements.txt
|-- src/
| |-- all code files
|-- results/
|-- figures/
|-- report.pdf
```

The repository must include:

- Instructions for installing dependencies and running the code
- Data source, reporting year, and acquisition instructions
- Descriptions of important processing steps
- Code for both Pandas and Spark analyses

- Code or procedures for checking result equivalence
- The performance experiment and its results

Do not upload sensitive credentials or unnecessarily duplicate large public datasets in the repository. Provide appropriate download instructions and document any instructor-provided files.

6.4 Deliverable 4: Presentation and Defense

Each group will present its project and answer questions. The presentation should cover:

1. The financial problem and research questions
2. Dataset characteristics and preparation
3. The most important analytical findings
4. The Pandas and Spark implementations
5. Whether the two implementations produced equivalent results
6. Performance and scalability results
7. Financial interpretation and limitations

Every group member should understand the methodology, implementation, and conclusions.

7 Suggested Project Timeline

Period	Stage	Expected progress
Week 1	Exploration and proposal	Understand the 2024 dataset, inspect data quality, formulate research questions, and submit the proposal.
Week 2	Data engineering	Clean and transform the data, document decisions, and prepare the analytical dataset.
Week 3	Descriptive analysis	Complete the main descriptive analyses and visualizations in Pandas and implement corresponding analyses in Spark.
Week 4	Comparison and scalability	Verify result equivalence, conduct performance experiments, and develop financial interpretations.
Week 5	Finalization and defense	Complete the report, code repository, presentation, and oral defense.

The instructor may adjust the timeline to match the course schedule.

8 Assessment Criteria

Criterion	Points	Assessment focus
Problem formulation (Proposal)	Counting towards midterm scores	Clear financial problem, well-defined research questions, and appropriate scope.

Criterion	Points	Assessment focus
Data engineering	20	Data understanding, quality assessment, justified cleaning, feature engineering, and reproducibility.
Descriptive and diagnostic analysis	25	Correct statistics, meaningful comparisons, relevant relationships, effective visualizations, and evidence-based findings.
Pandas–PySpark implementation and correctness comparison	20	Appropriate implementations, consistent definitions, verification of equivalent results, and investigation of discrepancies.
Performance and scalability experiment	10	Appropriate experimental design, reliable timing measurements, clear reporting, and thoughtful interpretation.
Financial interpretation	10	Financial relevance, sound reasoning, acknowledgment of limitations, and appropriately cautious conclusions.
Presentation and defense	15	Clear communication, logical organization, effective responses to questions, and demonstrated understanding by group members.
Total	**100**	

Assessment principle: Grades reflect analytical quality, technical correctness, result equivalence, performance evaluation, and financial reasoning rather than the number of charts, lines of code, or algorithms used.

Optional predictive analysis may earn up to ten bonus percentage points, subject to the overall score cap of 100. It is not required for full marks.

9 Submission Checklist

Before submitting, verify that your group has completed the following:

☐ The financial problem and research questions are clearly stated.

☐ The 2024 dataset source, schema, and variables are documented.

☐ Data quality issues and cleaning decisions are explained.

☐ Descriptive and diagnostic analyses answer the research questions.

☐ Visualizations are labeled and interpreted.

☐ Core analyses are implemented using both Pandas and PySpark.

☐ At least three important analytical outputs are compared.

☐ Any discrepancies between implementations are investigated.

☐ A performance and scalability experiment is reported.

☐ Financial implications and analytical limitations are discussed.

☐ The report, code, and execution instructions are complete.

☐ Every group member is prepared to defend the work.

10 Final Expectations

A successful project is not simply a collection of charts or a machine-learning model. It is a coherent analytical investigation that begins with a financial question, processes data reliably, uses both Pandas and Spark appropriately, verifies the consistency of the results, and ends with conclusions supported by evidence.

The strongest projects will demonstrate not only *what* the data reveal, but also *how reliably* the results were obtained, *how performance changes* with data volume, and *why* the findings matter for financial analysis.

End of Assignment